Workshop Equipment

Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Workshop tool/equipment

What it is used for

Automotive workshop equipment supports vehicle inspection, servicing, repair, lifting, measuring, and maintenance tasks. Correct selection, setup, and operation help technicians work efficiently while reducing injury and vehicle damage.

Main parts

- Equipment frame, base, or supporting structure
- Operating controls, handles, or switches
- Power supply, air supply, or hydraulic connections as applicable
- Guards, locks, and safety interlocks where fitted
- Adapters, attachments, and work-holding accessories
- Operating instructions and inspection records

Before you start

1. Identify the equipment type, intended task, rated capacity, and applicable operating instructions.
2. Inspect the frame, handles, guards, cables, hoses, connectors, and moving parts for damage.
3. Check that required inspection, testing, and calibration are current for the equipment type.
4. Confirm the work area is clean, adequately lit, and free from obstructions.
5. Verify that the correct attachment, adapter, and personal protective equipment are available.
6. Ensure the operator is trained and that power, air, or hydraulic supplies meet the equipment requirements.

How to use

1. Position the equipment on a stable surface or at the designated workshop station.
2. Secure the workpiece or vehicle using the approved holding and support points.
3. Connect the required power, air, or hydraulic supply according to the instructions.
4. Set the operating controls and select the correct accessory for the task.
5. Start or activate the equipment using the designated control.
6. Operate at the specified speed, pressure, load, or measurement range for the task.
7. Monitor the work for unusual noise, vibration, overheating, leakage, or unstable movement.
8. Stop the equipment immediately if a fault, unsafe condition, or abnormal result occurs.
9. Switch off and isolate energy sources before adjustments, tool changes, or cleaning.
10. Clean the equipment, inspect it for damage, and return it to its designated storage location.

⚠ Safety

- Use equipment only for its intended purpose and within its rated capacity.
- Wear task-appropriate personal protective equipment and secure loose clothing and hair.
- Never bypass guards, safety locks, or interlocks fitted to the equipment.
- Support raised vehicles and heavy components with approved stands or lifting devices.
- Isolate electrical, pneumatic, and hydraulic energy before servicing or clearing a blockage.
- Keep unauthorized people away from moving equipment and suspended loads.

Care & storage

- Clean the equipment after use and remove oil, grease, and debris safely.
- Inspect cables, hoses, fasteners, guards, and moving parts at the required intervals.
- Lubricate serviceable parts and replace worn accessories according to the manufacturer's instructions.
- Maintain inspection, calibration, repair, and preventive maintenance records where required.

Common mistakes to avoid

- Using equipment beyond its rated capacity can cause failure or serious injury.
- Selecting the wrong attachment can damage the workpiece and equipment.
- Ignoring damaged cables, hoses, guards, or locks creates avoidable hazards.
- Returning equipment to storage without cleaning or reporting defects can affect the next user.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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